

Due Wednesday, February 7 at 11:59 pm

- Homework 2 is an entirely written assignment; no coding involved.
- We prefer that you typeset your answers using $\text{\LaTeX}$ or other word processing software. If you haven't yet learned $\text{\LaTeX}$, one of the crown jewels of computer science, now is a good time! Neatly handwritten and scanned solutions will also be accepted.
- In all of the questions, **show your work**, not just the final answer.
- **Start early. This is a long assignment. Most of the material is prerequisite material not covered in lecture; you are responsible for finding resources to understand it.**

Deliverables:

1. Submit a PDF of your homework to the Gradescope assignment entitled "HW2 Write-Up". You may typeset your homework in $\text{\LaTeX}$ or Word (submit PDF format, **not** .doc/.docx format) or submit neatly handwritten and scanned solutions. **Please start each question on a new page.** If there are graphs, include those graphs in the correct sections. **Do not** put them in an appendix. We need each solution to be self-contained on pages of its own.
 - In your write-up, please state whom you had discussions with (not counting course staff) about the homework contents.
 - In your write-up, please copy the following statement and sign your signature next to it. (Mac Preview, PDF Expert, and FoxIt PDF Reader, among others, have tools to let you sign a PDF file.) We want to make it *extra* clear so that no one inadvertently cheats.
"I certify that all solutions are entirely in my own words and that I have not looked at another student's solutions. I have given credit to all external sources I consulted."

1 Identities and Inequalities with Expectation

For this exercise, the following identity might be useful: for a probability event A , $\mathbb{P}(A) = \mathbb{E}[\mathbf{1}\{A\}]$, where $\mathbf{1}\{\cdot\}$ is the indicator function.

$$\mathbf{1}_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{otherwise} \end{cases}$$

1. For any non-negative real-valued random variable X and constant $t > 0$, show that

$$\mathbb{P}(X \geq t) \leq \mathbb{E}[X]/t.$$

This is known as Markov's inequality.

Hint: show that for $a, b > 0$, $\mathbf{1}\{a \geq b\} \leq a/b$.

2. A common use for concentration inequalities (like Markov's inequality above) is to study the performance of a statistical estimator. For example, given a random real-valued variable Z with true mean μ and variance 1, we hope to estimate its mean using the regular estimator $\hat{\mu} = \sum_{i=1}^n Z_i/n$, where $\{Z_i\}_{1, \dots, n}$ are n Independent and identically distributed (i.i.d.) real values sampled from the distribution. Using Markov's inequality above, please show that

$$\mathbb{P}(|\hat{\mu} - \mu| \geq t) \leq \frac{1}{\sqrt{nt}}.$$

This tells us that the more samples we have, the better our estimator will be.

Hint: Use reverse Jensen's inequality to convert the mean of an absolute value into a mean of squares.

$$1. \quad \mathbb{P}(X \geq t) \leq \mathbb{E}[X]/t$$

$$\text{equals to } \mathbb{E}(\mathbf{1}\{X \geq t\}) \leq \mathbb{E}[X]/t$$

to show

prove that for $a, b > 0$ $\mathbf{1}\{a \geq b\} \leq a/b$

$$\mathbf{1}\{a \geq b\} = \begin{cases} 1 & a \geq b \\ 0 & a < b \end{cases}$$

$$\text{If } a \geq b \quad a/b \geq 1 \quad \mathbf{1}\{a \geq b\} = 1 \leq a/b$$

$$\text{If } a < b \quad 0 < a/b < 1 \quad \mathbf{1}\{a \geq b\} = 0 < a/b$$

$$\Rightarrow \mathbb{P}(X \geq t) \leq \mathbb{E}[X]/t$$

$$2. \quad \mu = \mathbb{E}[Z]$$

$$\sigma^2 = 1 = \text{Var}(Z) = \mathbb{E}[(Z - \mu)^2]$$

$$\hat{\mu} = \sum_{i=1}^n \frac{Z_i}{n}$$

$$\mathbb{P}(|\hat{\mu} - \mu| \geq t) \leq \mathbb{E}[|\hat{\mu} - \mu|]/t$$

Jensen inequality, convex φ , then

$$\varphi(\mathbb{E}(x)) \leq \mathbb{E}(\varphi(x))$$

$$\text{If } \varphi(x) = x^2$$

$$\Rightarrow (\mathbb{E}[Z])^2 \leq \mathbb{E}[Z^2]$$

$$\Rightarrow (\mathbb{E}[|\hat{\mu} - \mu|])^2 \leq \mathbb{E}[(\hat{\mu} - \mu)^2]$$

$$= \text{Var}[\hat{\mu}]$$

$$\text{Var}[\hat{\mu}] = \text{Var}\left[\frac{1}{n} \sum_{i=1}^n Z_i\right]$$

$$= \frac{1}{n^2} \sum_{i=1}^n \text{Var}[Z_i]$$

$$\text{Var}[Z_i] = \text{Var}[Z] = 1 \quad Z_i \text{ i.i.d.}$$

$$\Rightarrow \text{Var}[\hat{\mu}] = \frac{1}{n^2} \sum_{i=1}^n 1 = \frac{1}{n^2} \cdot n = \frac{1}{n}$$

$$\Rightarrow \mathbb{P}(|\hat{\mu} - \mu| \geq t) \leq \frac{1}{\sqrt{n}} / t = \frac{1}{\sqrt{nt}}$$

2 Probability Potpourri

- Recall that the covariance of two random variables X and Y is defined to be $\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$. For a multivariate random variable Z (i.e., each component of the vector Z is a random variable), we define the square covariance matrix Σ with entries $\Sigma_{ij} = \text{Cov}(Z_i, Z_j)$. Concisely, $\Sigma = \mathbb{E}[(Z - \mu)(Z - \mu)^T]$, where μ is the mean value of the column vector Z . Show that the covariance matrix is always positive semidefinite (PSD). You can use the definition of PSD matrices in Q3.2. Σ is a matrix $n \times n$.
- The probability that an archer hits her target when it is windy is 0.4; when it is not windy, her probability of hitting the target is 0.7. On any shot, the probability of a gust of wind is 0.3. Find the probability that
 - on a given shot there is a gust of wind and she hits her target.
 - she hits the target with her first shot.
 - she hits the target exactly once in two shots.
 - on an occasion when she missed, there was no gust of wind.
- An archery target is made of 3 concentric circles of radii $1/\sqrt{3}$, 1 and $\sqrt{3}$ feet. Arrows striking within the inner circle are awarded 4 points, arrows within the middle ring are awarded 3 points, and arrows within the outer ring are awarded 2 points. Shots outside the target are awarded 0 points.

Consider a random variable X , the distance of the strike from the center in feet, and let the probability density function of X be

$$f(x) = \begin{cases} \frac{2}{\pi(1+x^2)} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

What is the expected value of the score of a single strike?

$$1. \Sigma = \mathbb{E}[(Z - \mu)(Z - \mu)^T]$$

for any $x \in \mathbb{R}^n$

$$\begin{aligned} x^T \Sigma x &= x^T \mathbb{E}[(Z - \mu)(Z - \mu)^T] x \\ &= \mathbb{E} \left[x^T (Z - \mu)(Z - \mu)^T x \right] \\ &= \mathbb{E} \left[((Z - \mu)^T x)^T ((Z - \mu)^T x) \right] \\ &= \mathbb{E} \left[((Z - \mu)^T x)^2 \right] \geq 0 \end{aligned}$$

$\Rightarrow \Sigma$ is always positive-semidefinite

$$2. P(\text{windy}) = 0.3$$

$$P(1 | \text{not windy}) = 0.4$$

$$P(1 | \text{windy}) = 0.7$$

$$(i) P(1 \text{ and windy})$$

$$= P(1 | \text{windy}) P(\text{windy}) = 0.7 \times 0.3 = 0.21$$

$$(ii) P(1) = P(1 | \text{windy}) \cdot P(\text{windy})$$

$$+ P(1 | \text{not windy}) \cdot P(\text{not windy})$$

$$= 0.7 \times 0.3 + 0.4 \times 0.7 = 0.61$$

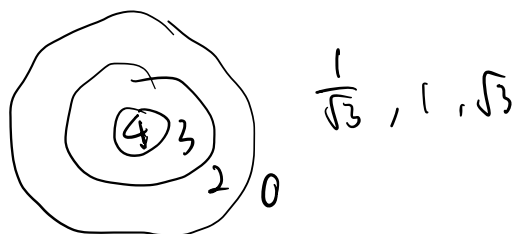
$$(iii) P(0) = 1 - P(1) = 1 - 0.61 = 0.39$$

$$P(1)P(0) + P(0)P(1) = 0.4758$$

$$(iv) P(\text{not windy} | 0) = \frac{P(0 | \text{not windy}) P(\text{not windy})}{P(0)}$$

$$\begin{aligned} &\text{missed is occasion} \\ &= \frac{0.3 \times 0.7}{0.39} = 0.5385 \end{aligned}$$

3.

 $\frac{1}{\sqrt{3}}, 1, \sqrt{3}$

$$E = E[\text{points} \cdot f(x)]$$

$$= \int_0^{\frac{1}{\sqrt{3}}} 4 \cdot \frac{2}{\pi(1+x^2)} dx + \int_{\frac{1}{\sqrt{3}}}^1 3 \cdot \frac{2}{\pi(1+x^2)} dx + \int_1^{\sqrt{3}} 2 \cdot \frac{2}{\pi(1+x^2)} dx$$

$$= 4 \cdot \frac{2}{\pi} \arctan x \Big|_0^{\frac{1}{\sqrt{3}}} + 3 \cdot \frac{2}{\pi} \arctan x \Big|_{\frac{1}{\sqrt{3}}}^1 + 2 \cdot \frac{2}{\pi} \arctan x \Big|_1^{\sqrt{3}}$$

$$= \frac{2}{\pi} \left[4 \cdot \left(\frac{\pi}{6} - 0 \right) + 3 \cdot \left(\frac{\pi}{4} - \frac{\pi}{6} \right) + 2 \cdot \left(\frac{\pi}{3} - \frac{\pi}{4} \right) \right]$$

$$\frac{2}{\pi} \left[\frac{2}{3} \pi + \frac{\pi}{4} + \frac{\pi}{6} \right] = \frac{13}{6}$$

3 Linear Algebra Review

1. First we review some basic concepts of rank. Recall that elementary matrix operations do not change a matrix's rank. Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$. Let I_n denote the $n \times n$ identity matrix.

- (a) Perform elementary row and column operations¹ to transform $\begin{bmatrix} I_n & 0 \\ 0 & AB \end{bmatrix}$ to $\begin{bmatrix} B & I_n \\ 0 & A \end{bmatrix}$.
- (b) Let's find lower and upper bounds on $\text{rank}(AB)$. Use part (a) to prove that $\text{rank } A + \text{rank } B - n \leq \text{rank}(AB)$. Then use what you know about the relationship between the column space (range) and/or row space of AB and the column/row spaces for A and B to argue that $\text{rank}(AB) \leq \min\{\text{rank } A, \text{rank } B\}$.
- (c) If a matrix A has rank r , then some $r \times r$ submatrix M of A has a nonzero determinant. Use this fact to show the standard facts that the dimension of A 's column space is at least r , and the dimension of A 's row space is at least r . (Note: You will not receive credit for other ways of showing those same things.)
- (d) It is a fact that $\text{rank}(A^T A) = \text{rank } A$; here's one way to see that. We've already seen in part (b) that $\text{rank}(A^T A) \leq \text{rank } A$. Suppose that $\text{rank}(A^T A)$ were strictly less than $\text{rank } A$. What would that tell us about the relationship between the column space of A and the null space of A^T ? What standard fact about the fundamental subspaces of A says that relationship is impossible?
- (e) Given a set of vectors $S \subseteq \mathbb{R}^n$, let $AS = \{Av : v \in S\}$ denote the subset of $\mathbb{R}^m$ found by applying A to every vector in S . In terms of the ideas of the column space (range) and row space of A : What is $A\mathbb{R}^n$, and why? (Hint: what are the definitions of column space and row space?) What is $A^T A\mathbb{R}^n$, and why? (Your answer to the latter should be purely in terms of the fundamental subspaces of A itself, not in terms of the fundamental subspaces of $A^T A$.)

2. Let $A \in \mathbb{R}^{n \times n}$ be a symmetric matrix. Prove equivalence between these three different definitions of positive semidefiniteness (PSD). Note that when we talk about PSD matrices, they are defined to be symmetric matrices. There are nonsymmetric matrices that exhibit PSD properties, like the first definition below, but not all three.

- (a) For all $x \in \mathbb{R}^n$, $x^T A x \geq 0$.
- (b) All the eigenvalues of A are nonnegative.
- (c) There exists a matrix $U \in \mathbb{R}^{n \times n}$ such that $A = UU^T$.

Positive semidefiniteness will be denoted as $A \geq 0$.

3. The Frobenius inner product between two matrices of the same dimensions $A, B \in \mathbb{R}^{m \times n}$ is

$$\langle A, B \rangle = \text{trace}(A^T B) = \sum_{i=1}^m \sum_{j=1}^n A_{ij} B_{ij},$$

¹If you're not familiar with these, <https://stattrek.com/matrix-algebra/elementary-operations> is a decent introduction.

where trace M denotes the *trace* of M , which you should look up if you don't already know it. (The norm is sometimes written $\langle A, B \rangle_F$ to be clear.) The Frobenius norm of a matrix is

$$\|A\|_F = \sqrt{\langle A, A \rangle} = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |A_{ij}|^2}.$$

Prove the following. The Cauchy-Schwarz inequality, the cyclic property of the trace, and the definitions in part 2 above may be helpful to you.

(a) $x^T A y = \langle A, x y^T \rangle$ for all $x \in \mathbb{R}^m, y \in \mathbb{R}^n, A \in \mathbb{R}^{m \times n}$.

(b) If A and B are symmetric PSD matrices, then $\text{trace}(AB) \geq 0$.

(c) If $A, B \in \mathbb{R}^{n \times n}$ are real symmetric matrices with $\lambda_{\max}(A) \geq 0$ and B being PSD, then

$$\langle A, B \rangle \leq \sqrt{n} \lambda_{\max}(A) \|B\|_F.$$

Hint: Construct a PSD matrix using $\lambda_{\max}(A)$

Let $A \in \mathbb{R}^{m \times n}$ be an arbitrary matrix. The maximum singular value of A is defined to be $\sigma_{\max}(A) = \sqrt{\lambda_{\max}(A^T A)} = \sqrt{\lambda_{\max}(A A^T)}$. Prove that

$$\sigma_{\max}(A) = \max_{\substack{u \in \mathbb{R}^m, v \in \mathbb{R}^n \\ \|u\|=1, \|v\|=1}} (u^T A v).$$

1. (a) $\checkmark$

$$\begin{bmatrix} I_n & 0 \\ 0 & AB \end{bmatrix} \rightarrow \begin{bmatrix} B & I_n \\ 0 & A \end{bmatrix}$$

$$\begin{bmatrix} I_n & 0 \\ 0 & AB \end{bmatrix} \begin{bmatrix} B & I \\ 0 & B^{-1} \end{bmatrix} = \begin{bmatrix} B & I_n \\ 0 & A \end{bmatrix}$$

first col $\leftarrow B \times \text{first col} + 0 \times \text{second col}$

second col $\leftarrow 1 \times \text{first col} + B^{-1} \times \text{second col}$

(b) $\checkmark$

$$A^{m \times n} \quad B^{n \times p}$$

$$(AB)^{m \times p}$$

$$\begin{aligned} n + \text{rank}(AB) &= \text{rank} \left(\begin{bmatrix} I_n & 0 \\ 0 & AB \end{bmatrix} \right) \\ &= \text{rank} \left(\begin{bmatrix} B & I_n \\ 0 & A \end{bmatrix} \right) \end{aligned}$$

$$\geq \text{rank}(A) + \text{rank}(B)$$

(c) $\checkmark$ M has $\text{rank} \neq 0$
its columns are linearly independent
rows

M has a rank of r

M is submatrix of $A \Rightarrow$

$\text{sub}(A) = M$ have r linearly independent cols rows

(d)

Suppose $\text{rank}(A^T A) < \text{rank}(A)$

exist v in column space of A
and null space of A^T

$\Rightarrow v$ in row space of A^T

row space orthogonal to null space

$\Rightarrow v \perp v$

and v is not zero $\Rightarrow$ Wrong.

(e)

AR^n is the column space of A .

Ax ($x \in \mathbb{R}^n$) is $A \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1 \begin{bmatrix} | \\ | \\ | \end{bmatrix} + \dots + x_n \begin{bmatrix} | \\ | \\ | \end{bmatrix}$

$$\text{rank}(AB) \leq \text{rank}(A)$$

$$\text{rank}(AB) \leq \text{rank}(B) \Rightarrow \text{rank}(AB)$$

$$A^T A \in \mathbb{R}^n \quad (A^T) \text{ A column space}$$



A row space

$$\text{rank}(A^T A) = \text{rank}(A)$$

2. (a) ✓ $\exists v$ as eigenvector

$$\rightarrow (b) \quad v^T A v = v^T \lambda v = \lambda v^T v = \lambda \|v\|^2 \geq 0.$$

$$\Rightarrow \lambda \geq 0.$$

(b) $\Rightarrow$ (c)

$$\lambda_i \geq 0.$$

$$A = V \Lambda V^T = V \Lambda' \Lambda'^T V^T$$

$$\Lambda' = \sqrt{\Lambda} = \begin{pmatrix} \sqrt{\lambda_1} & & \\ & \sqrt{\lambda_2} & \\ & & \ddots & \\ & & & \sqrt{\lambda_n} \end{pmatrix}$$

$$= U U^T$$

(c) $\Rightarrow$ (a) ✓

$$A = U U^T$$

$$\begin{aligned} x U U^T x^T &= (U^T x^T)^T (U^T x^T) \\ &= \|U^T x^T\|^2 \geq 0. \end{aligned}$$

$$3. (a) \checkmark \quad \langle A, x y^T \rangle = \text{trace}(A^T x y^T) = \text{trace}(y x^T A)$$

$$= \text{trace}(x^T A y).$$

$$= x^T A y.$$

(b) ✓ $\text{trace}(AB)$

$$\text{Let } \hat{A} = A^T \quad \hat{A} \text{ is PSD}$$

$$\Rightarrow \text{trace}(AB) = \text{trace}(\hat{A}^T B) = \langle \hat{A}, B \rangle$$

$$= \text{trace}(U U^T V V^T) = \text{trace}(V^T U U^T V)$$

$$A = U U^T \quad B = V V^T = \text{trace}(M^T M)$$

$$M = U^T V$$

$$= \langle M, M \rangle = \sum \sum |M_{ij}|^2 \geq 0.$$

(c) B is PSD

$$\lambda_B \geq 0.$$

$$\lambda_{\max}(A) \geq 0. \text{ let be } \lambda_{a0}$$

$$\text{Construct } M = \lambda_{a0} I - A$$

M all the eigenvalues > 0 .

$$\langle \lambda_{a0} I - A, B \rangle$$

$$\Rightarrow \lambda \langle I, B \rangle - \langle A, B \rangle \geq 0.$$

$$\langle A, B \rangle \leq \lambda_0 \langle I, B \rangle$$

$$= \lambda_0 \sum_{ij} B_{ij}.$$

$$\frac{\sum B_{ii}}{n} \leq \sqrt{\frac{\sum (B_{ii})^2}{n}}$$

$$\Rightarrow \sum B_{ii} \leq \sqrt{n} \sqrt{\sum B_{ii}^2} = \sqrt{n} \|B\|_F$$

$$4. \quad \sigma_{\max}(A) = \sqrt{\lambda_{\max}(A^T A)}$$

$$\sup v^T A^T A v = \sup v^T U^T \Lambda U v = \sup w^T \Lambda w = \sup \sum_i \lambda_i w_i^2$$

$A^T A$ symmetric PSD

$$w = Uv$$

$$v^T A^T A v \geq 0.$$

$$A^T A = U^T \Lambda U$$

$$\Lambda = \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{pmatrix}$$

$$\sup w^T \Lambda w = \sup \sum_i \lambda_i w_i^2$$

$$\sigma^2 = s$$

4 Matrix/Vector Calculus and Norms

1. Consider a 2×2 matrix A , written in full as $\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$, and two arbitrary 2-dimensional vectors x, y . Calculate the gradient of

$$\sin(A_{11}^2 + e^{A_{11}+A_{22}}) + x^\top A y$$

with respect to the matrix A .

Hint: The gradient has the same dimensions as A . Use the chain rule.

2. Aside from norms on vectors, we can also impose norms on matrices. Besides the Frobenius norm, the most common kind of norm on matrices is called the induced norm. Induced norms are defined as

$$\|A\|_p = \sup_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p}$$

where the notation $\|\cdot\|_p$ on the right-hand side denotes the vector ℓ_p -norm. Please give the closed-form (or the most simple) expressions for the following induced norms of $A \in \mathbb{R}^{m \times n}$.

- (a) $\|A\|_2$. (Hint: Similar to Question 3.4.)
 - (b) $\|A\|_\infty$.
3. (a) Let $\alpha = \sum_{i=1}^n y_i \ln \beta_i$ for $y, \beta \in \mathbb{R}^n$. What are the partial derivatives $\frac{\partial \alpha}{\partial \beta_i}$?
 (b) Let $\gamma = A\rho + b$ for $b \in \mathbb{R}^n, \rho \in \mathbb{R}^m, A \in \mathbb{R}^{n \times m}$. What are the partial derivatives $\frac{\partial \gamma_i}{\partial \rho_j}$?
 (c) Given $x \in \mathbb{R}^n, y \in \mathbb{R}^m, z \in \mathbb{R}^k$ and $y = f(x), f: \mathbb{R}^n \mapsto \mathbb{R}^m, z = g(y), g: \mathbb{R}^m \mapsto \mathbb{R}^k$. Please write the Jacobian $\frac{dz}{dx}$ as the product of two other matrices. What are these matrices?
 (d) Given $x \in \mathbb{R}^n, y, z \in \mathbb{R}^m$, and $y = f(x), z = g(x)$. Write the gradient $\nabla_x y^\top z$ in terms of y and z and some other terms.
 4. Let's apply the multivariate chain rule to a "simple" type of neural network called a *linear neural network*. They're not very powerful, as they can learn only linear regression functions or decision functions, but they're a good stepping stone for understanding more complicated neural networks. We are given an $n \times d$ *design matrix* X . Each row of X is a training point, so X represents n training points with d features each. We are also given an $n \times k$ matrix Y . Each row of Y is a set of k labels for the corresponding training point in X . Our goal is to learn a $k \times d$ matrix W of weights² such that

$$Y \approx XW^\top.$$

²The reason for the transpose on $W^\top$ is because we think in terms of applying W to an individual training point. Indeed, if $X_i \in \mathbb{R}^d$ and $Y_i \in \mathbb{R}^k$ respectively denote the i -th rows of X and Y transposed to be column vectors, then we can write $Y_i \approx WX_i$. For historical reasons, most papers in the literature use design matrices whose rows are sample points, rather than columns.

If n is larger than d , typically there is no W that achieves equality, so we seek an approximate answer. We do that by finding the matrix W that minimizes the *cost function*

$$\text{RSS}(W) = \|XW^\top - Y\|_F^2. \quad (1)$$

This is a classic *least-squares linear regression* problem; most of you have seen those before. But we are solving k linear regression problems simultaneously, which is why Y and W are matrices instead of vectors.

Linear neural networks. Instead of optimizing W over the space of $k \times d$ matrices directly, we write the W we seek as a product of multiple matrices. This parameterization is called a *linear neural network*.

$$W = \mu(W_L, W_{L-1}, \dots, W_2, W_1) = W_L W_{L-1} \cdots W_2 W_1.$$

Here, μ is called the *matrix multiplication map* (hence the Greek letter mu) and each W_j is a real-valued $d_j \times d_{j-1}$ matrix. Recall that W is a $k \times d$ matrix, so $d_L = k$ and $d_0 = d$. L is the number of *layers* of “connections” in the neural network. You can also think of the network as having $L + 1$ layers of units: $d_0 = d$ units in the *input layer*, d_1 units in the first *hidden layer*, d_{L-1} units in the last hidden layer, and $d_L = k$ units in the *output layer*.

We collect all the neural network’s weights in a *weight vector* $\theta = (W_L, W_{L-1}, \dots, W_1) \in \mathbb{R}^{d_\theta}$, where $d_\theta = d_L d_{L-1} + d_{L-1} d_{L-2} + \dots + d_1 d_0$ is the total number of real-valued weights in the network. Thus we can write $\mu(\theta)$ to mean $\mu(W_L, W_{L-1}, \dots, W_1)$. But you should imagine θ as a column vector: we take all the components of all the matrices $W_L, W_{L-1}, \dots, W_1$ and just write them all in one very long column vector. Given a fixed weight vector θ , the linear neural network takes an *input vector* $x \in \mathbb{R}^{d_0}$ and returns an *output vector* $y = W_L W_{L-1} \cdots W_2 W_1 x = \mu(\theta)x \in \mathbb{R}^{d_L}$.

Now our goal is to find a weight vector θ that minimizes the composition $\text{RSS} \circ \mu$ —that is, it minimizes the cost function

$$J(\theta) = \text{RSS}(\mu(\theta)).$$

We are substituting a linear neural network for W and optimizing the weights in θ instead of directly optimizing the components of W . This makes the optimization problem harder to solve, and you would never solve least-squares linear regression problems this way in practice; but again, it is a good exercise to work toward understanding the behavior of “real” neural networks in which μ is *not* a linear function.

We would like to use a gradient descent algorithm to find θ , so we will derive $\nabla_\theta J$ as follows.

- (a) The gradient $G = \nabla_W \text{RSS}(W)$ is a $k \times d$ matrix whose entries are $G_{ij} = \partial \text{RSS}(W) / \partial W_{ij}$, where $\text{RSS}(W)$ is defined by Equation (1). Write two explicit formulas for $\nabla_W \text{RSS}(W)$. First, derive a formula for each G_{ij} using summations, simplified as much as possible. Use that result to find a simple formula for $\nabla_W \text{RSS}(W)$ in matrix notation with no summations.

- (b) Directional derivatives are closely related to gradients. The notation $\text{RSS}'_{\Delta W}(W)$ denotes the directional derivative of $\text{RSS}(W)$ in the direction ΔW , and the notation $\mu'_{\Delta\theta}(\theta)$ denotes the directional derivative of $\mu(\theta)$ in the direction $\Delta\theta$.³ Informally speaking, the directional derivative $\text{RSS}'_{\Delta W}(W)$ tells us how much $\text{RSS}(W)$ changes if we increase W by an infinitesimal displacement $\Delta W \in \mathbb{R}^{k \times d}$. (However, any ΔW we can actually specify is not actually infinitesimal; $\text{RSS}'_{\Delta W}(W)$ is a local linearization of the relationship between W and $\text{RSS}(W)$ at W . To a physicist, $\text{RSS}'_{\Delta W}(W)$ tells us the initial velocity of change of $\text{RSS}(W)$ if we start changing W with velocity ΔW .)

Show how to write $\text{RSS}'_{\Delta W}(W)$ as a Frobenius inner product of two matrices, one related to part (a).

- (c) In principle, we could take the gradient $\nabla_{\theta} \mu(\theta)$, but we would need a 3D array to express it! As I don't know a nice way to write it, we'll jump directly to writing the directional derivative $\mu'_{\Delta\theta}(\theta)$. Here, $\Delta\theta \in \mathbb{R}^{d_{\theta}}$ is a weight vector whose matrices we will write $\Delta\theta = (\Delta W_L, \Delta W_{L-1}, \dots, \Delta W_1)$. Show that

$$\mu'_{\Delta\theta}(\theta) = \sum_{j=1}^L W_{>j} \Delta W_j W_{<j}$$

where $W_{>j} = W_L W_{L-1} \cdots W_{j+1}$, $W_{<j} = W_{j-1} W_{j-2} \cdots W_1$, and we use the convention that $W_{>L}$ is the $d_L \times d_L$ identity matrix and $W_{<1}$ is the $d_0 \times d_0$ identity matrix.

Hint: although μ is not a linear function of θ , μ is linear in any *single* W_j ; and any directional derivative of the form $\mu'_{\Delta\theta}(\theta)$ is linear in $\Delta\theta$ (for a fixed θ).

- (d) Recall the chain rule for scalar functions, $\frac{d}{dx} f(g(x))|_{x=x_0} = \frac{d}{dy} f(y)|_{y=g(x_0)} \cdot \frac{d}{dx} g(x)|_{x=x_0}$. There is a multivariate version of the chain rule, which we hope you remember from some class you've taken, and the multivariate chain rule can be used to chain directional derivatives. Write out the chain rule that expresses the directional derivative $J'_{\Delta\theta}(\theta)|_{\theta=\theta_0}$ by composing your directional derivatives for RSS and μ , evaluated at a weight vector θ_0 . (Just write the pure form of the chain rule without substituting the values of those directional derivatives; we'll substitute the values in the next part.)
- (e) Now substitute the values you derived in parts (b) and (c) into your expression for $J'_{\Delta\theta}(\theta)$ and use it to show that

$$\begin{aligned} \nabla_{\theta} J(\theta) = & (2(\mu(\theta) X^{\top} - Y^{\top}) X W_{<L}^{\top}, \dots, \\ & 2W_{>j}^{\top} (\mu(\theta) X^{\top} - Y^{\top}) X W_{<j}^{\top}, \dots, \\ & 2W_{>1}^{\top} (\mu(\theta) X^{\top} - Y^{\top}) X). \end{aligned}$$

This gradient is a vector in $\mathbb{R}^{d_{\theta}}$ written in the same format as $(W_L, \dots, W_j, \dots, W_1)$. Note that the values $W_{>j}$ and $W_{<j}$ here depend on θ .

Hint: you might find the cyclic property of the trace handy.

³“ ΔW ” and “ $\Delta\theta$ ” are just variable names that remind us to think of these as small displacements of W or θ ; the Greek letter delta is not an operator nor a separate variable.

5. (Optional bonus question worth 1 point. This question contains knowledge that goes beyond the scope of this course, and is intended as an exercise to really make you comfortable with matrix calculus). Consider a differentiable function $f : \mathbb{R}^n \mapsto \mathbb{R}$. Suppose this function admits a unique global optimum $x^* \in \mathbb{R}^n$. Suppose that for some spherical region $X = \{x \mid \|x - x^*\|^2 \leq D\}$ around x^* for some constant D , the Hessian matrix H of the function $f(x)$ is PSD and its maximum eigenvalue is 1. Prove that

$$f(x) - f(x^*) \leq \frac{D}{2}$$

for every $x \in X$. *Hint: Look up Taylor's Theorem with Remainder. Use Mean Value Theorem on the second order term instead of the first order term, which is what is usually done.*

$$1. \quad \left(\nabla_A (x^T A y) \right)_{ij} = \left[\frac{\partial \sum x_j A_{ji} y_i}{\partial A} \right]_{ij} = \frac{\partial \sum x_j A_{ji} y_i}{\partial A_{ji}} = x_j y_i = y_i x_j = (y x^T)_{ij}$$

$$\Rightarrow \nabla_A (x^T A y) = (y x^T)^T = x y^T \quad \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} [y_1 \ y_2] = \begin{bmatrix} x_1 y_1 & x_1 y_2 \\ x_2 y_1 & x_2 y_2 \end{bmatrix} \quad \textcircled{2}$$

$$\left(\frac{\partial (\sin(A_{11}^2 + e^{A_{11} + A_{22}}))}{\partial A} \right)_{ij}$$

$$= \frac{\partial \sin(A_{11}^2 + e^{A_{11} + A_{22}})}{\partial A}$$

$$\frac{\partial}{\partial A_{ji}} \sin(A_{11}^2 + e^{A_{11} + A_{22}}) = \cos(A_{11}^2 + e^{A_{11} + A_{22}}) (2A_{11} + e^{A_{11} + A_{22}})$$

$$\frac{\partial}{\partial A_{22}} \sin(A_{11}^2 + e^{A_{11} + A_{22}}) = \cos(A_{11}^2 + e^{A_{11} + A_{22}}) e^{A_{11} + A_{22}}$$

$$\nabla_A \nabla_A^T = \begin{bmatrix} \bigcirc & 0 \\ 0 & \bigcirc \end{bmatrix}^T = \begin{bmatrix} \bigcirc & 0 \\ 0 & \bigcirc \end{bmatrix} \quad \textcircled{1}$$

$$\textcircled{1} + \textcircled{2}$$

$$2. (a) \|A\|_2 = \sup_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2} = \sup_{\|x\|_2=1} \|Ax\|_2$$

$$\|x\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

$$\text{SVD of } A: A = U \Sigma V^T$$

$$\begin{aligned} \sup \frac{\|Ax\|_2}{\|x\|_2} &= \sup \frac{\|U \Sigma V^T x\|_2}{\|x\|_2} \\ &= \sup \frac{\|\Sigma V^T x\|_2}{\|x\|_2} \quad V^T x = y \\ &\quad V y. \end{aligned}$$

$$\begin{aligned} (b) \sup \frac{\|Ax\|_\infty}{\|x\|_\infty} &= \sup \frac{\max_i \left| \sum_{j=1}^n a_{ij} x_j \right|}{\max_i |x_i|} \\ &= \sup_{x \neq 0} \frac{\max_i \sum_{j=1}^n |a_{ij} x_j|}{\max_i |x_i|} \\ \|Ax\|_\infty &= \begin{array}{l} \text{but } \sum a_{ii} x_i \\ \text{also } \sum a_{ii} x_i \end{array} \\ &= \sup_{x \neq 0} \frac{\max_i \sum_{j=1}^n |a_{ij}| |x_j|}{\max_i |x_i|} \end{aligned}$$

$$|x_j| / \max_i |x_i| \leq 1 \quad j=1, 2, \dots, n.$$

$$= \max_j \sum_{i=1}^n |a_{ij}|$$

$$= \max_i (|a_{i1}| + |a_{i2}| + \dots + |a_{in}|)$$

$$3. (a) \alpha = \sum_{i=1}^n y_i \ln \beta_i$$

$$\frac{\partial \alpha}{\partial \beta_i} = \frac{y_i}{\beta_i}$$

$$(b) r = Ap + b \quad r_i = \sum_j A_{ij} p_j + b_i$$

$$\frac{\partial r_i}{\partial p_j} = \frac{\partial (\sum_j A_{ij} p_j + b_i)}{\partial p_j} = A_{ij}.$$

$$(c) y = f(x) \\ z = g(y).$$

$$\left(\frac{dz}{dx} \right)_{ij} = \frac{dz_i}{dx_j} = \sum_{k=1}^m \frac{\partial z_i}{\partial y_k} \frac{\partial y_k}{\partial x_j}$$

$$= \sum_{k=1}^m g'_{ik} f'_{kj}$$

$$= (g' \circ f')_{ij}$$

$$\Rightarrow \frac{dz}{dx} = g' \circ f' \\ = \frac{dz}{dy} \cdot \frac{dy}{dx}$$

$$\text{Jacob}_g \cdot \text{Jacob}_f$$

$$(d) y = f(x) \quad z = g(x)$$

$$\left(\frac{\partial y^T z}{\partial x} \right)_i \quad \boxed{\quad}$$

$$= \frac{\partial y^T z}{\partial x_i}$$

$$= \frac{\partial \sum_i f(x_i) g(x_i)}{\partial x_i}$$

$$= g(x_i) \frac{\partial f}{\partial x} \Big|_{x=x_i} + f(x_i) \frac{\partial g}{\partial x} \Big|_{x=x_i}$$

$$= f'(x_i) g(x_i) + g'(x_i) f(x_i)$$

$$= [f'(x)^T g(x) + g'(x)^T f(x)]_i$$

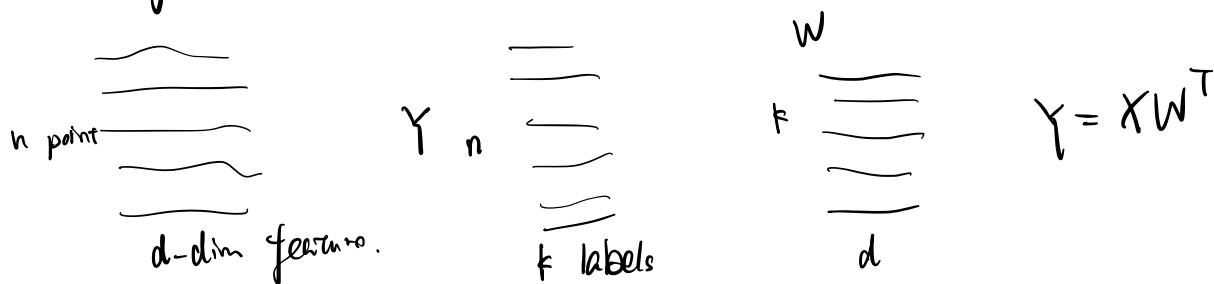
$$(\quad)$$

$$\frac{y^T z}{x} = ((f'(x)^T g(x) + f(x)^T g'(x))^T)$$

$$= f'(x)^T g(x) + g'(x)^T f(x)$$

$$= \left(\frac{dy}{dx} \right)^T z + \left(\frac{dz}{dx} \right)^T y$$

4. design matrix X $n \times d$



$$\|A\|_F^2 = \sum_i \sum_j |A_{ij}|^2$$

$$RSS(W) = \|XW^T - Y\|_F^2$$

$$W \text{ is } k \times d \quad \nabla_W RSS(W)$$

$$(a) G_{ij} = \frac{\partial RSS(W)}{\partial W_{ij}} = \frac{\partial}{\partial W_{ij}} (\|XW^T - Y\|_F^2)$$

$$= \frac{\partial \sum_i \sum_j (XW^T - Y)_{ij}^2}{\partial W_{ij}} = \frac{\partial}{\partial W_{ij}} \text{trace}[(XW^T - Y)^T (XW^T - Y)]$$

$$= \frac{\partial}{\partial W_{ij}} \sum_m (WX^T - Y^T)(XW^T - Y)_{mm}$$

$$= \frac{\partial}{\partial W_{ij}} \left(\sum_m (WX^T X W^T)_{mm} - \sum_m (Y^T X W^T)_{mm} - \sum_m (WX^T Y)_{mm} + \sum_m (Y^T Y)_{mm} \right)$$

$$= \frac{\partial}{\partial W_{ij}} \left(\sum_m (WX^T X W^T)_{mm} - 2 \sum_m (WX^T Y)_{mm} \right)$$

$$= \frac{\partial}{\partial W_{ij}} \left(\sum_m \sum_k \sum_l \sum_s W_{mk} X_{lk} X_{ls} W_{ms} - 2 \sum_m \sum_k \sum_l W_{mk} X_{lk} Y_{lm} \right)$$

$$= \textcircled{1} \quad m=i \quad k=j \quad \sum_l \sum_s X_{lj} X_{ls} W_{is}$$

$$+ \textcircled{2} \quad m=i \quad s=j \quad \sum_k \sum_l W_{ik} X_{lk} X_{lj}$$

$$- 2 \textcircled{3} \quad m=i \quad k=j \quad \sum_l X_{lj} Y_{li}$$

$$= 2 \sum_l \sum_k W_{ik} X_{lk} X_{lj} - 2 \sum_l Y_{li} X_{lj}$$

$$\Rightarrow \nabla_W \text{RSS} = 2WX^T X - 2Y^T X = 2(WX^T - Y^T)X$$

$$(b) \quad \text{RSS}'_{\Delta W}(W) = \underset{\substack{\uparrow \\ \text{链式}}}{\Delta W} \cdot \nabla_W \text{RSS}(W) = \langle G, \Delta W \rangle_F$$

$$= \langle 2WX^T X - 2Y^T X, \Delta W \rangle_F$$

$$(c) \quad \Delta \theta = (\Delta W_L, \Delta W_{L-1}, \dots, \Delta W_1)$$

$$\nabla_W \mu(W) = \nabla_W W_L W_{L-1} \dots W_2 W_1$$

$$\nabla_W \cdot \Delta W =$$

$$\mu(W + \Delta W) = \mu(W_L + \Delta W_L, W_{L-1} + \Delta W_{L-1}, \dots, W_1 + \Delta W_1)$$

$$= \prod_{j=1}^L (W_j + \Delta W_j)$$

不考虑 $(\Delta W_j)^2$ 及其高阶项

$$= \underbrace{\prod_{j=1}^L W_j} + \sum_{j=1}^L \Delta W_j \prod_{\substack{i=1 \\ i \neq j}}^L W_i$$

$$= \mu(W) + \sum_{j=1}^L \Delta W_j \prod_{\substack{i=1 \\ i \neq j}}^L W_i$$

$$\begin{aligned} \mu'_{\Delta \theta}(\theta) &= \mu(W + \Delta W) - \mu(W) = \sum_{j=1}^L \Delta W_j \prod_{\substack{i=1 \\ i \neq j}}^L W_i \\ &= \sum_{j=1}^L W_{>j} \Delta W_j W_{<j} \end{aligned}$$

$$(d) \quad J(\theta) = \text{RSS}(\mu(\theta))$$

$$J'_{\Delta \theta}(\theta) \big|_{\theta=\theta_0} = \frac{d \text{RSS}(y)}{dy} \bigg|_{y=\mu(\theta_0)} \cdot \underset{\substack{\downarrow \\ \text{链式}}}{\frac{d\mu}{d\theta}} \bigg|_{\theta=\theta_0}$$

$$= \langle \nabla_W \text{RSS}(\mu(\theta_0)), \mu'_{\Delta \theta}(\theta_0) \rangle_F$$

$$\mu(\theta) = \mu(W_L, W_{L-1}, \dots, W_1) = W_L W_{L-1} \dots W_1 = W$$

$$\begin{aligned}
(e) \quad J'_{\Delta\theta}(\theta) &= \left\langle 2\mu(\theta) X^T X - 2Y^T X, \sum_{j=1}^L W_{>j} \Delta W_j W_{<j} \right\rangle \\
&= \text{trace} \left(\left(\sum_{j=1}^L W_{>j} \Delta W_j W_{<j} \right)^T (2\mu(\theta) X^T - Y^T) X \right) \\
&= \text{trace} \left(\sum_{j=1}^L W_{<j}^T \Delta W_j^T W_{>j}^T (2\mu(\theta) X^T - Y^T) X \right) \\
&= \text{trace} \left(\sum_{j=1}^L \left[2W_{>j}^T (\mu(\theta) X^T - Y^T) X W_{<j}^T \right] \Delta W_j^T \right) \\
&= \sum_{j=1}^L \text{trace} (M_j \Delta W_j^T) \\
&= \sum_{j=1}^L \langle M_j, \Delta W_j \rangle
\end{aligned}$$

$$\begin{aligned}
J'_{\Delta\theta}(\theta) &= \nabla_{\theta} J(\theta) \underline{\Delta\theta} \\
\Delta\theta &= (\Delta W_L, \Delta W_{L-1}, \dots, \Delta W_1) \\
\Rightarrow \nabla_{\theta} J(\theta) &= (M_L, M_{L-1}, \dots, M_1)
\end{aligned}$$

$$\begin{aligned}
5. \quad f(x) &\approx f(x^*) + \nabla f(x^*)^T (x - x^*) + \frac{1}{2} (x - x^*)^T \nabla^2 f(\xi) (x - x^*) \\
&\quad \uparrow \\
&\quad \text{Hessian} \\
\xi \in (x, x^*) &= tx + (1-t)x^* \\
0 \leq t &\leq 1.
\end{aligned}$$

$$\Rightarrow \xi \in X$$

$$f(x) - f(x^*) = \nabla f(x^*)^T (x - x^*) + \frac{1}{2} (x - x^*)^T \nabla^2 f(\xi) (x - x^*)$$

global optimum: $\nabla f(x^*) = 0$.

$$\nabla^2 f(\xi) \text{ max eigenvalue: } 1.$$

$$\begin{aligned}
\frac{1}{2} (x - x^*)^T \nabla^2 f(\xi) (x - x^*) &\leq \frac{1}{2} \lambda_{\max} (x - x^*)^T (x - x^*) \\
&\quad \underbrace{\quad}_{U \Sigma V^T} \\
&\leq \lambda_{\max} U I_n V^T = \lambda_{\max} I_n = I_n \\
&\quad \downarrow \\
&= \frac{1}{2} \cdot 1 \cdot \|x - x^*\|^2 = \frac{D}{2}
\end{aligned}$$

5 Properties of the Normal Distribution (Gaussians)

1. Prove that $\mathbb{E}[e^{\lambda X}] = e^{\sigma^2 \lambda^2 / 2}$, where $\lambda \in \mathbb{R}$ is a constant, and $X \sim \mathcal{N}(0, \sigma^2)$. As a function of λ , $M_X(\lambda) = \mathbb{E}[e^{\lambda X}]$ is also known as the *moment-generating function*.
2. *Concentration inequalities* are inequalities that place upper bounds on the likelihood that a random variable X is far away from its mean μ , written $\mathbb{P}(|X - \mu| \geq t)$, with a falling exponential function ae^{-bt^2} having constants $a, b > 0$. Such inequalities imply that X is very likely to be close to its mean μ . To make a tight bound, we want a to be as small and b to be as large as possible.

For $t > 0$ and $X \sim \mathcal{N}(0, \sigma^2)$, prove that $\mathbb{P}(X \geq t) \leq \exp(-t^2/2\sigma^2)$, then show that $\mathbb{P}(|X| \geq t) \leq 2 \exp(-t^2/2\sigma^2)$.

Hint: Consider using Markov's inequality and the result from Question 5.1.

3. Let $X_1, \dots, X_n \sim \mathcal{N}(0, \sigma^2)$ be i.i.d. (independent and identically distributed). Find a concentration inequality, similar to Question 5.2, for the average of n Gaussians: $\mathbb{P}(\frac{1}{n} \sum_{i=1}^n X_i \geq t)$? What happens as $n \rightarrow \infty$?

Hint: Without proof, use the fact that linear combinations of i.i.d. Gaussian-distributed variables are also Gaussian-distributed. Be warned that summing two Gaussian variables does not mean that you can sum their probability density functions (no no no!).

4. Let vectors $u, v \in \mathbb{R}^n$ be constant (i.e., not random) and orthogonal (i.e., $\langle u, v \rangle = u \cdot v = 0$). Let $X = (X_1, \dots, X_n)$ be a vector of n i.i.d. standard Gaussians, $X_i \sim \mathcal{N}(0, 1), \forall i \in [n]$. Let $u_x = \langle u, X \rangle$ and $v_x = \langle v, X \rangle$. Are u_x and v_x independent? Explain. If $X_1, \dots, X_n$ are independently but not identically distributed, say $X_i \sim \mathcal{N}(0, i)$, does the answer change? *Hint:* two jointly normal random variables are independent if and only if they are uncorrelated.

$$1. \quad X \sim \mathcal{N}(0, \sigma^2) \Rightarrow f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x^2}{2\sigma^2}\right)$$

$$\mathbb{E}[e^{\lambda x}] = \int_{-\infty}^{\infty} e^{\lambda x} f_X(x) dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{\lambda x} e^{-\frac{x^2}{2\sigma^2}} dx$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x - \sigma^2 \lambda)^2}{2\sigma^2} + \frac{1}{2}\sigma^2 \lambda^2} dx$$

$$= e^{\frac{1}{2}\sigma^2 \lambda^2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x - \sigma^2 \lambda)^2}{2\sigma^2}} dx$$

$$= e^{\sigma^2 \lambda^2 / 2}$$

$$\begin{aligned} \frac{\sigma^2 \lambda^2}{2} - \lambda t &= -\frac{t^2}{2\sigma^2} \\ \Rightarrow \sigma^2 \lambda^2 - 2t\lambda + \frac{t^2}{\sigma^2} &= 0 \\ \Rightarrow (\sigma\lambda - \frac{t}{\sigma}) &= 0 \\ \Rightarrow \lambda &= \frac{t}{\sigma^2} \end{aligned}$$

2. Markov's inequality: $\mathbb{P}(X \geq t) \leq \mathbb{E}[X]/t$. for $X \geq 0$

$$\mathbb{P}(X \geq t) = \mathbb{P}(e^{\lambda x} \geq e^{\lambda t}) \leq \mathbb{E}[e^{\lambda x}] / e^{\lambda t}$$

$$= e^{\sigma^2 \lambda^2 / 2 - \lambda t}$$

$$\mathbb{P}(|X| \geq t) = \mathbb{P}(X \geq t) + \mathbb{P}(X \leq -t) = 2\mathbb{P}(X \geq t).$$

$$\begin{aligned} \Rightarrow \mathbb{P}(X \geq t) &\leq e^{\frac{\sigma^2 \lambda^2}{2} - \lambda t} \\ &= e^{-\frac{t^2}{2\sigma^2}} \end{aligned}$$

$$3. \quad \tilde{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$\begin{aligned} \text{Var}(\tilde{X}) &= \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \text{Var}\left(\sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) \\ &= \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{\sigma^2}{n} \end{aligned}$$

$$\mu(\tilde{X}) = \frac{1}{n} \sum_{i=1}^n \mu(X_i) = 0.$$

$$\Rightarrow \tilde{X} \sim (0, \frac{\sigma^2}{n}).$$

$$P(\tilde{X} \geq t) \leq \exp\left(-t^2 / 2\left(\frac{\sigma^2}{n}\right)\right) = \exp\left(-nt^2 / 2\sigma^2\right)$$

$$4. \quad \text{Cov}(u_x, v_x) = E[u_x v_x] - E[u_x] E[v_x]$$

$$\begin{aligned} E[u_x] &= E[\langle u, X \rangle] = E\left[\text{trace}\left(\sum_{i=1}^n u_i X_i\right)\right] = E\left[\sum_{i=1}^n u_i \cdot \text{trace}(X_i)\right] \\ &= \sum_{i=1}^n u_i \cdot \text{trace}(E[X_i]) \end{aligned}$$

$$X \sim N(0, I) \Rightarrow E[X_i] = 0 \quad i=1, 2, \dots, n$$

$$\Rightarrow E[u_x] = 0 \quad \text{同理} \quad E[v_x] = 0$$

$$E[u_x v_x] = E[\langle u, X \rangle \langle v, X \rangle] = E\left[\left(\sum_{i=1}^n u_i X_i\right) \left(\sum_{j=1}^n v_j X_j\right)\right]$$

$$= E\left[\sum_{i=1}^n \sum_{j=1}^n u_i v_j X_i X_j\right] = \sum_{i=1}^n \sum_{j=1}^n u_i v_j E[X_i X_j]$$

$$i \neq j \text{ 时 } E[X_i X_j] = E[X_i] \cdot E[X_j] = 0$$

$$i = j \text{ 时 } E[u_x v_x] = \sum_{i=1}^n u_i v_i E[X_i^2] = \left(\sum_{i=1}^n u_i v_i\right) E[X^2] = \langle u, v \rangle E[X^2] = 0$$

$$\text{若 } X_i \sim N(0, i) \quad i=1, 2, \dots, n$$

$$E[X_i^2] : \quad \text{Var}[X_i] = E[X_i^2] - (E[X_i])^2 \Rightarrow i = E[X_i^2]$$

$$\Rightarrow E[u_x v_x] = \sum_{i=1}^n u_i v_i E[X_i^2] = \sum_{i=1}^n i u_i v_i \neq 0$$

$$\Rightarrow \text{Cov}(u_x, v_x) \neq 0$$

6 The Multivariate Normal Distribution

The multivariate normal distribution with mean $\mu \in \mathbb{R}^d$ and positive definite covariance matrix $\Sigma \in \mathbb{R}^{d \times d}$, denoted $\mathcal{N}(\mu, \Sigma)$, has the probability density function

$$f(x; \mu, \Sigma) = \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \exp\left(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right).$$

Here $|\Sigma|$ denotes the determinant of Σ . You may use the following facts without proof.

- The volume under the normal PDF is 1.

$$\int_{\mathbb{R}^d} f(x) dx = \int_{\mathbb{R}^d} \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \exp\left\{-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right\} dx = 1.$$

- The change-of-variables formula for integrals: let f be a smooth function from $\mathbb{R}^d \rightarrow \mathbb{R}$, let $A \in \mathbb{R}^{d \times d}$ be an invertible matrix, and let $b \in \mathbb{R}^d$ be a vector. Then, performing the change of variables $x \mapsto z = Ax + b$,

$$x = A^{-1}(z - b) = A^{-1}z - A^{-1}b \quad dx = |A^{-1}| dz$$

$$\int_{\mathbb{R}^d} f(x) dx = \int_{\mathbb{R}^d} f(A^{-1}z - A^{-1}b) |A^{-1}| dz.$$

1. Let $X \sim \mathcal{N}(\mu, \Sigma)$. Use a suitable change of variables to show that $\mathbb{E}[X] = \mu$.
2. Use a suitable change of variables to show that $\text{Var}(X) = \Sigma$, where the variance of a vector-valued random variable X is

$$\text{Var}(X) = \text{Cov}(X, X) = \mathbb{E}[(X - \mu)(X - \mu)^\top] = \mathbb{E}[XX^\top] - \mu\mu^\top.$$

Hints: Every symmetric, positive semidefinite matrix Σ has a symmetric, positive definite square root $\Sigma^{1/2}$ such that $\Sigma = \Sigma^{1/2}\Sigma^{1/2}$. Note that Σ and $\Sigma^{1/2}$ are invertible. After the change of variables, you will have to find another variance $\text{Var}(Z)$; if you've chosen the right change of variables, you can solve that by solving the integral for each diagonal component of $\text{Var}(Z)$ and a second integral for each off-diagonal component. The diagonal components will require integration by parts.

$$1. X \sim \mathcal{N}(\mu, \Sigma)$$

$$f(x) = \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \exp\left[-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right]$$

$$\Sigma^{-1} = U^\top \Lambda U = U^\top \sqrt{\Lambda}^\top \sqrt{\Lambda} U$$

$$(x - \mu)^\top U^\top \sqrt{\Lambda}^\top \sqrt{\Lambda} U (x - \mu)$$

$$= [\sqrt{\Lambda} U (x - \mu)]^\top [\sqrt{\Lambda} U (x - \mu)]$$

$$\frac{1}{2} A = \sqrt{\Lambda} U \quad b = -\sqrt{\Lambda} U \mu \quad -A^{-1} b = +\mu$$

$$z = \sqrt{\Lambda} U (x - \mu)$$

$$\Rightarrow z = Ax + b$$

$$\mathbb{E}[X] = \mathbb{E}[A^{-1}(z - b)]$$

$$= \int A^{-1}(z - b) f(A^{-1}z - A^{-1}b) |A^{-1}| dz$$

$$= \int A^{-1}z \underbrace{f(A^{-1}z - A^{-1}b)}_{\text{odd}} |A^{-1}| dz$$

$$+ \int -A^{-1}b f(A^{-1}z - A^{-1}b) |A^{-1}| dz$$

$$= 0 + (-A^{-1}b) \cdot 1 \quad \int = 1$$

$$= -A^{-1}b = +\mu.$$

$$2. \quad \text{Var}(X) = \text{Cov}(X, X) = E[(X - \mu)(X - \mu)^T] = E[XX^T] - \mu\mu^T$$

$$A = \Sigma^{-1/2} \quad z = A(x - \mu) \Rightarrow x - \mu = A^{-1}z$$

$$|\Sigma| = |\Sigma^{1/2}|^2 = 1/|A|^2$$

$$E[(x - \mu)(x - \mu)^T] = \int (x - \mu)(x - \mu)^T \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu)\right)$$

$$= \int A^{-1} z z^T A^{-1T} \frac{1}{\sqrt{(2\pi)^d}} |A| \exp\left(-\frac{1}{2} z^T \cancel{A^{-1T} A^T A A^{-1}} z\right)$$

$$= \int A^{-1} z z^T A^{-1T} \frac{1}{\sqrt{(2\pi)^d}} |A| \exp\left(-\frac{1}{2} z^T z\right)$$

$$= E[A^{-1} z z^T A^{-1T}] = A^{-1} E[z z^T] A^{-1T}$$

$$= \Sigma^{-1/2} \text{Var}(z) \Sigma^{-1/2T} = \Sigma^{-1/2} I \Sigma^{-1/2T} = I = \text{Var}(X).$$

$$X \sim N(\mu, \Sigma) \Rightarrow z = A(x - \mu) = \Sigma^{-1/2}(x - \mu) \sim N(0, I)$$

7 Gradient Descent

Consider the optimization problem $\min_{x \in \mathbb{R}^n} \frac{1}{2} x^T A x - b^T x$, where $A \in \mathbb{R}^{n \times n}$ is a PSD matrix with $0 < \lambda_{\min}(A) \leq \lambda_{\max}(A) < 1$.

1. Find the optimizer x^* (in closed form).
2. Solving a linear system directly using Gaussian elimination takes $O(n^3)$ time, which may be wasteful if the matrix A is sparse. For this reason, we will use gradient descent to compute an approximation to the optimal point x^* . Write down the update rule for gradient descent with a step size of 1 (i.e., taking a step whose length is the length of the gradient).
3. Show that the iterates $x^{(k)}$ satisfy the recursion $x^{(k)} - x^* = (I - A)(x^{(k-1)} - x^*)$.
4. Using Question 3.4, prove $\|Ax\|_2 \leq \lambda_{\max}(A)\|x\|_2$.
Hint: Use the fact that, if λ is an eigenvalue of A , then λ^2 is an eigenvalue of A^2 .
5. Using the previous two parts, show that for some $0 < \rho < 1$,

$$\|x^{(k)} - x^*\|_2 \leq \rho \|x^{(k-1)} - x^*\|_2.$$

6. Let $x^{(0)} \in \mathbb{R}^n$ be the starting value for our gradient descent iterations. If we want a solution $x^{(k)}$ that is $\epsilon > 0$ close to x^* , i.e. $\|x^{(k)} - x^*\|_2 \leq \epsilon$, then how many iterations of gradient descent should we perform? In other words, how large should k be? Give your answer in terms of ρ , $\|x^{(0)} - x^*\|_2$, and ϵ .

$$1. \quad \min \frac{1}{2} x^T A x - b^T x$$

$$x^* \text{ 处有 } \nabla_x \left(\frac{1}{2} x^T A x - b^T x \right) = 0.$$

$$\left[\frac{\partial}{\partial x} \left(\frac{1}{2} x^T A x - b^T x \right) \right]_i$$

$$= \frac{\partial}{\partial x_i} \left(\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n x_i A_{ij} x_j - \sum_{i=1}^n b_i x_i \right)$$

$$= \frac{1}{2} \sum_{j=1}^n A_{ij} x_j + \frac{1}{2} \sum_{j=1}^n x_j A_{ji} - b_i$$

$$= \frac{1}{2} (Ax)_i + \frac{1}{2} (A^T x)_i - b_i$$

$$= \left[\frac{1}{2} (A + A^T) x - b \right]_i \Rightarrow \nabla_x \left(\frac{1}{2} x^T A x - b^T x \right) = Ax - b$$

$$\frac{1}{2} (A + A^T) x = b$$

$$A \text{ is PSD matrix } \Rightarrow A = A^T$$

$$\Rightarrow Ax^* = b \Rightarrow x^* = A^{-1} b$$

$$2. \quad x \leftarrow x - \nabla_x \left(\frac{1}{2} x^T A x - b^T x \right)$$

$$= x - (Ax - b)$$

$$3. \quad x^{(k)} - x^*$$

$$= x^{(k-1)} - (Ax^{(k-1)} - b) - x^*$$

$$= Ix^{(k-1)} - Ax^{(k-1)} + I \cdot b - A^{-1}b$$

$$= (I - A)x^{(k-1)} - (I - A^{-1})b$$

$$= (I - A)x^{(k-1)} + (A^{-1} - I)b$$

$$= (I - A)x^{(k-1)} + (I - A)A^{-1}b$$

$$= (I - A)(x^{(k-1)} - x^*).$$

4. if λ is eigenvalue of A

then λ^2 is eigenvalue of $A^2 = A^T A$

$$\|Ax\|_2^2 = (Ax)^T (Ax) = x^T A^T A x$$

$$A^T A = U \Sigma_2 V \quad \Sigma_2 = \begin{bmatrix} \lambda_1^2 & & \\ & \lambda_2^2 & \\ & & \ddots \\ & & & \lambda_n^2 \end{bmatrix}$$

$$x^T U^T = u^T \cdot \|x\|_2$$

$$V x = V \cdot \|x\|_2$$

$$\begin{aligned} x^T A^T A x &= u^T \Sigma_2 V \|x\|_2^2 \leq \sqrt{\lambda_{\max}(\Sigma_2^T \Sigma_2)} \|x\|_2^2 \\ &= \sqrt{\lambda_{\max}^2 \cdot \lambda_{\max}^2} \|x\|_2^2 = \lambda_{\max}^2 \|x\|_2^2 \end{aligned}$$

$$\Rightarrow \|Ax\|_2 \leq \lambda_{\max} \|x\|_2$$

5. $\|x^{(k)} - x^*\|_2 = \|(I - A)(x^{(k-1)} - x^*)\|_2$

$I - A$ is also PSD

$$\leq \lambda_{\max}(I - A) \|x^{(k-1)} - x^*\|_2$$

$$\lambda_{\max}(I - A) = I - \lambda_{\min}(A) = I - \lambda_{\min} \equiv \rho \quad \#.$$

6. $\|x^{(k)} - x^*\|_2 \leq \rho \|x^{(k-1)} - x^*\|_2 \leq \rho^2 \|x^{(k-2)} - x^*\|_2$
 $\leq \dots \leq \rho^k \|x^{(0)} - x^*\|_2 \leq \epsilon.$

$$\Rightarrow \rho^k \leq \frac{\epsilon}{\|x^{(0)} - x^*\|_2}$$

$$\Rightarrow k_{\max} = \left\lceil \log_{\rho} \frac{\epsilon}{\|x^{(0)} - x^*\|_2} \right\rceil$$